



An approach for upgrading biomass and pyrolysis product quality using a combination of aqueous phase bio-oil washing and torrefaction pretreatment



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HIGHLIGHTS

- A novel approach was performed to upgrade the quality of biomass pyrolysis products.
- Aqueous phase bio-oil washing pretreatment removed metals from the sample.
- Torrefaction reduced the oxygen but increased the metal content of cotton stalk.
- 3D FTIR results were consistent with the pyrolysis stages of the TG/DTG curve.
- Combined pretreatment reduced water and acid and increased phenol content of bio-oil.

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ABSTRACT

Bio-oil undergoes phase separation because of poor stability. Practical application of aqueous phase bio-oil is challenging. In this study, a novel approach that combines aqueous phase bio-oil washing and torrefaction pretreatment was used to upgrade the biomass and pyrolysis product quality. The effects of individual and combined pretreatments on cotton stalk pyrolysis were studied using TG-FTIR and a fixed bed reactor. The results showed that the aqueous phase bio-oil washing pretreatment removed metals and resolved the two pyrolysis peaks in the DTG curve. Importantly, it increased the bio-oil yield and improved the pyrolysis product quality. For example, the water and acid content of bio-oil decreased significantly along with an increase in phenol formation, and the heating value of non-condensable gases improved, and these were more pronounced when combined with torrefaction pretreatment. Therefore, the combined pretreatment is a promising method, which would contribute to the development of polygeneration pyrolysis technology.

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1. Introduction

Fast pyrolysis of biomass for bio-oil production is one of the most promising technologies used for the utilization of biomass resources. Although bio-oil can be directly used for industrial furnace combustion, it is still a low-grade liquid fuel because of its inferior quality compared to those of traditional fossil fuels (Lehto et al., 2014; No, 2014). The low quality of the bio-oil is due to its poor stability, phase separation propensity, high water

and acid content, low heating value, and complicated composition. In fact, low quality of biomass pyrolysis products directly results from low quality of biomass raw materials (Chen et al., 2017). Therefore, an effective approach is required to enhance the quality of biomass raw materials, which can thereby improve the quality of the pyrolysis products.

It should be noted that bio-oil, and in particular, the aqueous phase bio-oil produced after bio-oil phase separation, contains a high amount of water and organic acids (Chen et al., 2016a). Since phase separation of bio-oil is almost inevitable, it may be worthwhile to use the aqueous and oil phase bio-oil fractions separately. Aqueous phase bio-oil has high water and acid contents and small amounts of ketones and phenols, which hinder its use as a fuel or

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as a source of raw chemical materials (Chen et al., 2014c). Previous studies have shown that organic acids can effectively remove metallic species from biomass, and the phenolic substances in bio-oils have also been helpful in removing metallic species (Karnowo et al., 2014; Oudenhoven et al., 2013; Pecha et al., 2015; Zhang and Xiong, 2016). In this case, washing the raw biomass with aqueous phase bio-oil was considered as a promising pretreatment method to improve the quality of the biomass and consequently the quality of the biomass pyrolysis products.

Additionally, torrefaction pretreatment at 200–300 °C in an inert gas atmosphere has also been an effective method for biomass pretreatment (Chen et al., 2015b). Torrefaction pretreatment significantly removes oxygen from biomass, increases the heating value, and improves the grinding properties of biomass (Arias et al., 2008; Chen et al., 2015c,d; Gil et al., 2015). More importantly, it efficiently reduces the acid and water content while increasing the heating value of the bio-oil (Zhang et al., 2015; Zheng et al., 2012, 2013). However, torrefaction pretreatment has certain limitations such as a reduced yield of bio-oil because of an increase in ash and metal content in the torrefied biomass and catalytic cracking of some high-value compounds to small molecule compounds (Chen et al., 2017; Zheng et al., 2015b). The combination of aqueous phase bio-oil washing and torrefaction pretreatments can harness the advantages of each and improve the quality of both the biomass and the pyrolysis products.

Thermogravimetry-Fourier transform infrared spectroscopy (TG-FTIR) has been widely used to study the characteristics of biomass pyrolysis. It can be used to detect weight loss during biomass pyrolysis, and most importantly, it can be used to detect the emission characteristics of the volatile components from pyrolysis in real time (Chen et al., 2014b). In addition, a fixed bed reactor has also been widely used to collect the solid, liquid, and gaseous products of biomass pyrolysis for separate analysis. However, to our knowledge, investigations using TG-FTIR and a fixed bed reactor to assess the effects of aqueous phase bio-oil washing and torrefaction pretreatments on pyrolysis have not been reported.

In this study, an approach that combines aqueous phase bio-oil washing and torrefaction pretreatment was used to upgrade cotton stalk. The effects of the aqueous phase bio-oil washing pretreatment, the torrefaction pretreatment, and the combined pretreatment on the quality of the pyrolysis products of cotton stalk were compared. The goal of this work was to provide data highlighting the application of aqueous phase bio-oil in the polygeneration pyrolysis of biomass.

2. Materials and methods

2.1. Materials

Cotton stalks were used as the experimental material. It was screened into particles with a size 40–60 mesh and dried at 105 °C for 6 h.

2.2. Aqueous phase bio-oil washing

During storage, phase separation of the bio-oil occurred. The originally homogeneous bio-oil gradually divided into two fractions: the aqueous phase and the oil phase. In this study, the aqueous phase bio-oil used to wash the cotton stalks was obtained from the pyrolysis of cotton stalks at 500 °C. The details of the pyrolysis process were as reported in the literature (Chen et al., 2014a). The water content of the aqueous phase bio-oil was 73.18 wt.%. Certain organic compounds with relatively high peak areas were detected using gas chromatography/mass spectrometry (GC/MS). These included acetic acid, hydroxyacetone, 1-hydroxy-2-butanone,

2-cyclopentenone, 1,2-cyclopentanedione, furfural, 2-furan-methanol, 2,3-dihydro-benzofuran, phenol, 2-methyl-phenol, 2-methoxy-phenol, 1,2-benzenediol, 2,6-dimethoxyphenol, and guaiacol. The peak area% of the acids, ketones, furans, and phenols were recorded as 32.28%, 16.39%, 9.46%, and 21.73%, respectively.

For the washing pretreatment, a cotton stalk sample (10 g) was added to a beaker containing 200 mL of aqueous phase bio-oil and stirred at 60 °C for 2 h. After washing, the cotton stalk sample was filtered, rinsed with deionized water until reaching a neutral pH, and dried at 105 °C for 12 h.

2.3. Torrefaction

Torrefaction experiments of cotton stalk samples were performed at 260 °C for 30 min using a tube furnace (OTL1200, Nanjing University Instrument Plant, China). Nitrogen was used as the carrier gas with a flow rate of 300 mL/min. Details of the torrefaction process were as reported in our previous study (Chen et al., 2017).

2.4. Sample labels

The dried cotton stalk was denoted as CS. The stalk pretreated with aqueous phase bio-oil washing was denoted as W-CS, and the one pretreated with torrefaction was denoted as T-CS. The stalk pretreated sequentially with aqueous phase bio-oil washing and torrefaction was denoted as TW-CS.

2.5. TG-FTIR experiments

TG-FTIR analysis of CS, W-CS, T-CS, and TW-CS were performed using a thermogravimetric analyzer (TGA Q500, TA Instrument, USA) connected to a Fourier-transform infrared spectrometer (Nicolet 6700, Thermo Scientific, USA). The amount of cotton stalk used for each experiment was 20 mg. The temperature was increased from 50 °C to 650 °C at a rate of 10 °C/min. The nitrogen flow rate was 70 mL/min. The IR spectra were recorded at 4000–400 cm^{−1} with a resolution of 1 cm^{−1}, and 3D FTIR spectrograms were obtained after the experiments.

2.6. Pyrolysis

Pyrolysis of each CS, W-CS, T-CS, and TW-CS was performed using a fixed bed pyrolysis reactor with 5 g of sample at 500 °C for 15 min with a nitrogen flow rate of 300 mL/min for each experiment. The detailed experimental process was described in a previous study (Chen et al., 2015a). After the experiment, solid (biochar), liquid (bio-oil), and gaseous (non-condensable gases) products were collected. The mass yields of biochar and bio-oil were obtained by mass measurements and that of the non-condensable gases was obtained from the difference of masses.

2.7. Product characterization

The non-condensable gases were detected using a gas chromatography analyzer (GC-TCD 7890, Shanghai Tianmei, China), and the high heating value (HHV) was calculated approximately by summing each gas concentration with its corresponding HHV. The water content of the bio-oil was determined via Karl-Fischer titration, and the HHV was measured using an adiabatic oxygen bomb calorimeter (XRY-1A, Changji Geological Instruments, China). The pH of the bio-oil was measured using a digital pH meter (PHS-3C, Shanghai leici, China), and the organic components were determined by gas chromatography coupled to a mass spectrometer (GC/MS 7890A/5975C, Agilent Company, USA). Proximate analysis of the solid samples was performed according to the Chi-

nese National Standards GB/T28731-2012. Ultimate analysis was performed using an elemental analyzer (Vario macro cube, Elementar, Germany), and oxygen was estimated by the difference. The metallic species were determined using an inductively coupled plasma optical emission spectrometer (ICP-OES, Optima 7300 DV, PerkinElmer, USA). The method to measure the HHV of the solid samples was same as that used for the bio-oil. Additionally, biochar was added to de-ionized water in a mass ratio of 1:20 for measuring pH.

2.8. Reproducibility of experiments

The analytical experiments were repeated three times, and the data was averaged. Most results were uniform between each run, and the relative error was generally less than 5%. The standard deviation of the data is also given.

3. Results and discussion

3.1. Properties of original and pretreated cotton stalk

3.1.1. Effects of pretreatments on fuel properties

The proximate analysis, ultimate analysis, and HHV of the original and the pretreated cotton stalks are shown in Table 1. The ash content decreased from 5.54 wt.% for CS to 4.34 wt.% for W-CS and increased to 7.39 wt.% for T-CS. This indicated that the aqueous phase bio-oil washing pretreatment clearly removed the ash from the samples. However, the torrefaction pretreatment increased the ash content, which was mainly because of hemicellulose decomposition, accompanied by small amounts of cellulose and lignin decomposition. These decompositions released large amounts of CO₂ and CO and some condensable gaseous products such as water and acetic acid during the torrefaction pretreatment (Chen et al., 2014d; Chen and Kuo, 2011; Ma et al., 2015; Zheng et al., 2015a). Thus, the remaining ash in the cotton stalk increased the ash content after the torrefaction pretreatment. The presence of ash negatively affects the quality and subsequent use of biomass. Interestingly, the ash content of TW-CS was less than that of T-CS, and even less than that of CS. This implies that sequential pretreatments using aqueous phase bio-oil washing and torrefaction also reduced the ash content, and this offset the negative effect of the increased ash content caused by torrefaction.

The volatile and fixed carbon content of the sample increased slightly after washing with aqueous phase bio-oil, resulting in a minor increase in the HHV of the sample. The effect of the torrefaction pretreatment on the fuel quality of the samples was greater than that of the aqueous phase bio-oil washing pretreatment. Compared to CS and W-CS, the volatile content of T-CS decreased, and the fixed carbon content increased. Meanwhile, a drastic increase in carbon content and a decrease in oxygen content also reduced the molar ratios of H/C and O/C, and increased the HHV of the sample. As shown in Table 1, the combination of the aqueous phase bio-oil washing and the torrefaction pretreatments had an important impact on the fuel quality of the sample. Generally, the fuel property values of TW-CS were between those of W-CS and T-CS.

The molar ratios of H/C and O/C in TW-CS were lower than those in CS, and these are crucial for the subsequent use of the biomass because a higher oxygen content is responsible for the low quality of the raw biomass and its products.

3.1.2. Effects of pretreatments on metallic species

Fig. 1 shows the contents of metallic species in the samples. The K content was higher than that of other metals. After pretreatment with aqueous phase bio-oil washing, the content of different metallic species was decreased to different extents. K and Na were most effectively removed, with removal rates of 85–90%. The removal rates of Ca, Mg, and Fe were each more than 60%. These results indicated that the aqueous phase bio-oil washing pretreatment effectively removed the metallic species, and this was mainly attributed to the dissolution, precipitation, and removal of the metallic species by water and acetic acid. However, the metallic species remained in the cotton stalk during the torrefaction process, and thus, the metallic species contents in T-CS were higher than those in CS and W-CS. These results were consistent with the changes in the ash contents of the samples.

The content of metallic species in TW-CS was lower because most of the metallic species were removed by washing, and this was beneficial for biomass pyrolysis and bio-oil production. In general, the combination of aqueous phase bio-oil washing and torrefaction pretreatments improved the fuel properties and also dramatically reduced the metallic species contents of cotton stalks.

3.2. Effects of pretreatments on pyrolysis behavior

3.2.1. TG analysis

The TG/DTG curves of the samples before and after the pretreatments are shown in Fig. 2. TG analysis showed that all of the samples went through three stages during pyrolysis. The first stage was a micro-weight loss process. This involved the evaporation of water and a slight removal of volatile compounds. The weight loss stage started at 200 °C where the DTG curve showed an obvious weight loss peak, which could be due to the decomposition of cellulose, hemicellulose, and lignin. In the third stage, above a temperature of approximately 450 °C, the weight loss was slower, and the weight loss rate was relatively small.

Pretreatment with both aqueous phase bio-oil washing and torrefaction had a significant effect on the pyrolysis behavior of cotton stalks. Each pretreatment increased the temperature of the maximum loss rate of the cotton stalk and shifted the entire TG/DTG curve to the right compared to that observed for CS. Aqueous phase bio-oil washing reduced the final residue mass, whereas torrefaction had the opposite effect. Previous studies have shown that metallic species have a strong catalytic effect on biomass pyrolysis and result in the production of less bio-oil and more biochar (Mourant et al., 2011; Zhang and Xiong, 2016). In the present study, pretreatment with aqueous bio-oil washing removed large amounts of metallic species in the samples, and therefore reduced the catalytic effects of these metals on the pyrolysis process. For the torrefaction pretreatment, large-scale decomposition of hemicellulose occurred because of its poor thermal stability.

Table 1
The fuel properties of original and pretreated cotton stalk.

Sample	Proximate analysis (wt.%, db)			Ultimate analysis (wt.%, db)				HHV (MJ/kg)
	Volatile	Fixed Carbon	Ash	[C]	[H]	[O]	[N]	
CS	76.93 ± 1.13	17.53 ± 0.35	5.54 ± 0.11	45.16 ± 0.91	6.28 ± 0.13	42.35 ± 0.76	0.67 ± 0.01	16.83 ± 0.36
W-CS	78.29 ± 0.89	17.37 ± 0.32	4.34 ± 0.07	46.93 ± 0.85	6.15 ± 0.14	41.93 ± 0.79	0.65 ± 0.01	17.36 ± 0.35
T-CS	63.37 ± 0.83	29.22 ± 0.25	7.39 ± 0.11	52.36 ± 0.91	5.58 ± 0.13	34.05 ± 0.63	0.62 ± 0.01	19.25 ± 0.36
TW-CS	69.44 ± 0.87	25.73 ± 0.43	4.83 ± 0.10	52.38 ± 0.75	5.64 ± 0.12	36.52 ± 0.68	0.63 ± 0.01	20.39 ± 0.38

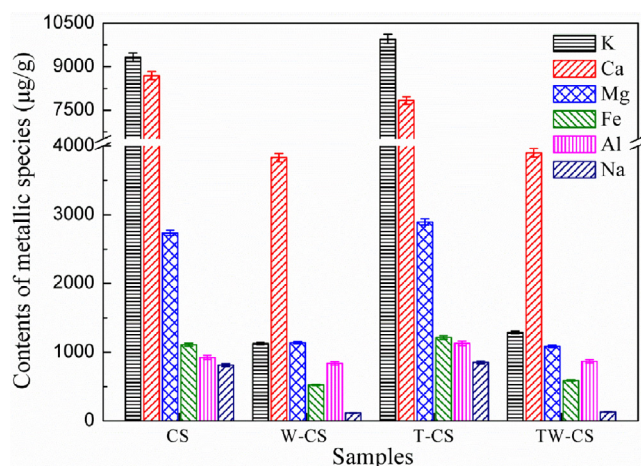


Fig. 1. The contents of metallic species in CS, W-CS, T-CS, and TW-CS.

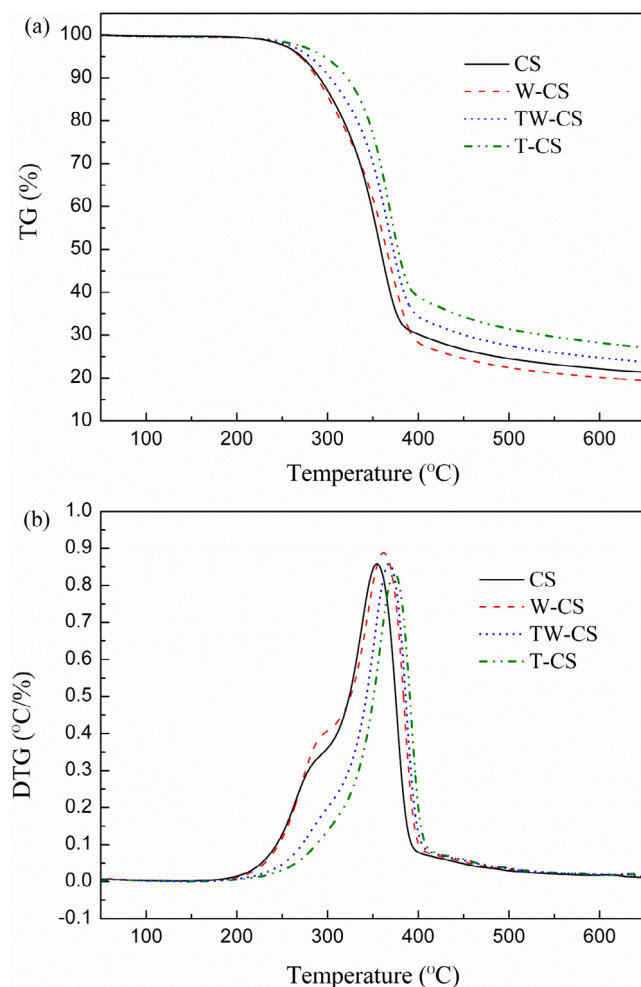


Fig. 2. TG (a) and DTG (b) curves of CS, W-CS, T-CS, and TW-CS.

However, since lignin is more thermally stable, the lignin content in the torrefied cotton stalk increased. Cellulose and hemicellulose contributed significantly to the formation of the liquid pyrolysis product (bio-oil), whereas lignin contributed mainly to the formation of the solid pyrolysis product (biochar) (Xin et al., 2013). Thus, the final residue mass of the sample increased after the torrefaction pretreatment.

The aqueous phase bio-oil washing and torrefaction pretreatments affected the pyrolysis peaks of the cotton stalks. In the DTG curve, there were two pyrolysis peaks. The first shoulder peak was primarily due to the pyrolysis of hemicellulose, and the second shoulder peak was due to cellulose pyrolysis (Ma et al., 2015). Aqueous phase bio-oil washing pretreatment increased the height of the first peak and improved the resolution of the two pyrolysis peaks. These observations might be related to the effects that the acidic compounds present in the aqueous phase bio-oil had on the composition and structure of the samples. Previous studies have shown that pretreatment with acid washing increased the amorphous and loose structures of cellulose and hemicellulose, and this eased biomass pyrolysis and promoted the production of several small molecule compounds (Álvarez et al., 2004; Branca et al., 2011). Meanwhile, acid washing removed most of the metallic species, which led to a higher pyrolysis temperature range of cellulose (Szabó et al., 1996). However, both the torrefaction pretreatment alone and the combined pretreatments led to the disappearance of the first shoulder peak. This result was attributed to the decomposition of large amounts of hemicellulose during torrefaction, which decreased the pyrolysis peak of hemicellulose.

3.2.2. 3D FTIR analysis

The three-dimensional FTIR spectrograms of the pyrolysis volatiles of CS, W-CS, T-CS, and TW-CS are shown in Fig. 3. The various FTIR spectra with time had a linear relationship with pyrolysis temperature. Clearly, there were three pyrolysis stages (Fig. 3), which were consistent with the TG/DTG curves shown in Fig. 2.

As seen in Fig. 3, in the first pyrolysis stage, almost no compounds were found, because the weight loss of the samples was small. Then pyrolysis volatiles were gradually generated and numerous compounds were detected using FTIR, and the typical compounds were identified by their characteristic absorbances (Chen et al., 2014b). The bands at $4000\text{--}3400\text{ cm}^{-1}$, $3050\text{--}2650\text{ cm}^{-1}$, $2400\text{--}2240\text{ cm}^{-1}$, and $2230\text{--}2000\text{ cm}^{-1}$ represent H_2O , CH_4 , CO_2 , and CO , respectively. The $\text{C}=\text{O}$ stretching absorbance was obvious between 1880 and 1620 cm^{-1} and represented the aldehyde, ketone, and acid organic components. The bands between 1600 and 400 cm^{-1} were very complex because of the characteristic absorbances of several organic compounds. The strongest spectral intensity of pyrolysis volatiles was found at 35–38 min, which corresponded to the mass loss peak observed in the DTG curve. With an increase in pyrolysis time and pyrolysis temperature, the amount of pyrolysis volatiles decreased, and the spectral intensity gradually decreased.

A shoulder peak was observed in the 3D FTIR spectrograms of CS, and it was also obvious in the 3D FTIR spectrograms of W-CS. However, it was not present in the spectrograms for T-CS and TW-CS because of the decomposition of hemicellulose during torrefaction. Generally, the spectral intensities of the pyrolysis volatiles of W-CS and TW-CS were larger than those of CS and T-CS, respectively, suggesting that the pyrolysis of cotton stalks was facilitated by the aqueous phase bio-oil washing pretreatment.

3.3. Distribution of pyrolysis products

Temperature is a key pyrolysis parameter that has a significant impact on the distribution and composition of products. Generally, $500\text{ }^{\circ}\text{C}$ is the temperature reported for maximum bio-oil yield (Chen et al., 2016b). The effects of pretreatments on the product yields obtained from pyrolysis at $500\text{ }^{\circ}\text{C}$ are shown in Fig. 4. In Fig. 4(a), the product yields of CS are based on the original mass of CS, and the product yields of the pretreated samples are based on the masses of the pretreated samples.

Compared to the product yields from the pyrolysis of CS, a higher bio-oil yield and a lower biochar yield were observed for

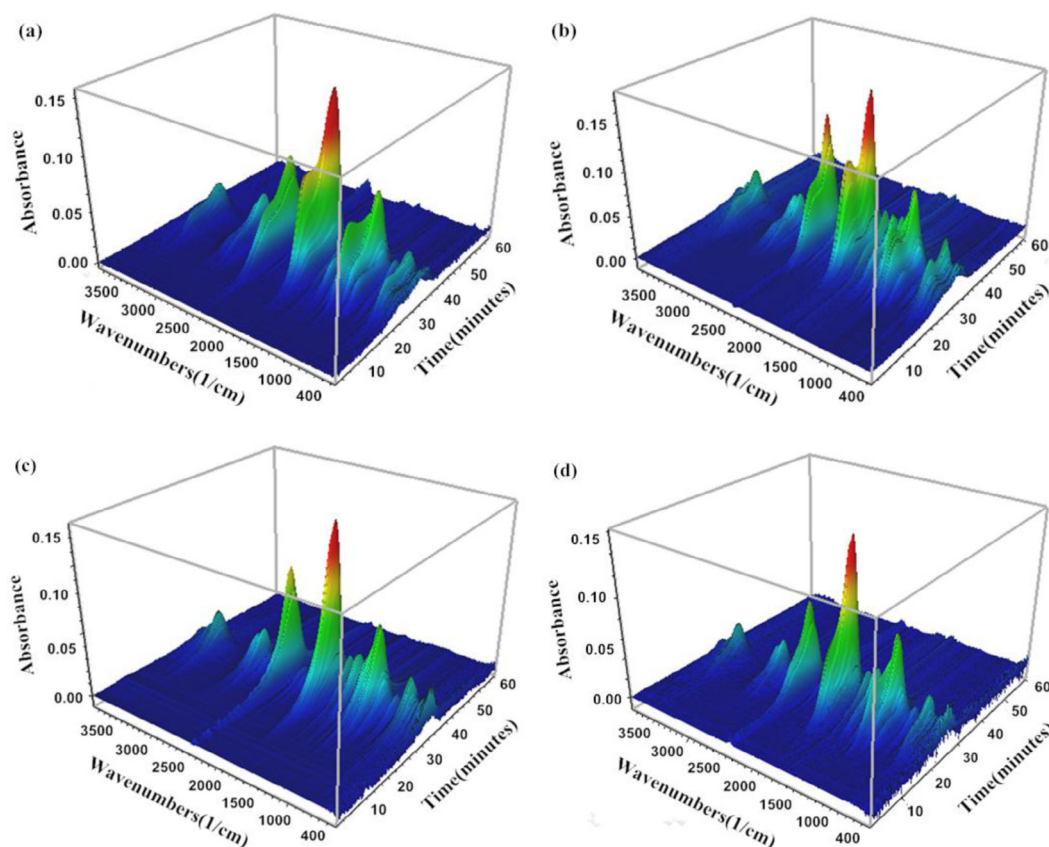


Fig. 3. 3D FTIR spectrograms of pyrolysis volatiles of the samples: CS (a), W-CS (b), T-CS (c), and TW-CS (d).

W-CS, indicating that the aqueous phase bio-oil washing pretreatment promoted the formation of bio-oil, while it inhibited the formation of biochar from biomass pyrolysis. These phenomena mainly resulted from the catalytic effect of the metallic species. The aqueous phase bio-oil washing pretreatment removed a large amount of the metallic species, and thus altered the degradation pathways of pyrolysis to increase the bio-oil yield. However, compared to the aqueous phase bio-oil washing pretreatment, the torrefaction pretreatment had the opposite effect on the distribution of bio-oil and biochar. Compared to the product yields of CS and W-CS, the bio-oil yields of T-CS decreased, while those of biochar and the non-condensable gases increased. This was mainly attributed to three factors. First, the metallic species were enriched in the cotton stalks after the torrefaction pretreatment. Second, a large amount of hemicellulose decomposed during the torrefaction pretreatment, and this decreased the formation of bio-oil in the subsequent pyrolysis procedure of the cotton stalks. The third factor was the carbonization and cross-linking of cellulose during torrefaction which contributed to the formation of biochar from fast pyrolysis (Chaiwat et al., 2008). Generally, the product yields of TW-CS were between those of W-CS and T-CS, indicating that the combined pretreatments included the individual influences of the aqueous phase bio-oil washing pretreatment and the torrefaction pretreatment.

It should be noted that there was a mass reduction in the cotton stalk after the pretreatments. In Fig. 4(b), the product yields of all of the samples are based on the original mass of CS. The bio-oil yield of W-CS was basically unchanged compared to that of CS, whereas the yields of biochar and the non-condensable gases decreased (Fig. 4(b)). In contrast, the torrefaction pretreatment greatly reduced the production of bio-oil, while the yields of biochar and the non-condensable gases only slightly changed. This

further confirmed that torrefaction decreases the bio-oil yield of biomass pyrolysis. For the combined pretreatment, all of the product yields were reduced, indicating that more feedstock was required to meet the production target. Although introducing the aqueous phase bio-oil washing pretreatment to the biomass pyrolysis decreased the output of products to different extents, the quantity loss could be compensated for if the quality of the products were improved.

3.4. Properties of the pyrolysis products

3.4.1. Non-condensable gases

In the following sections, the pyrolysis product yields were based on the pretreated samples. The gaseous products of the fast pyrolysis of cotton stalks were mainly CO, CO₂, CH₄, and H₂. Some C₂₊ gases, including C₂H₂, C₂H₄, C₂H₆, C₃H₆, and C₃H₈, were also detected but in very low volume fractions. The volume fractions of the non-condensable gases produced during cotton stalk pyrolysis are shown in Table 2. The volume fraction of CO₂ was the highest among the non-condensable gases produced by the pyrolysis of CS, followed by CO, CH₄, and H₂. As shown in Table 2, the aqueous phase bio-oil washing and torrefaction pretreatments affected the volume fractions of the non-condensable gases. Compared to the volume fractions of the non-condensable gases produced by the pyrolysis of CS, the volume fraction of CO₂ from W-CS decreased, while that of CH₄ slightly increased. The volume fractions of CO and H₂ changed negligibly. This might be a result of the catalytic effects of the metallic species during pyrolysis, whereby the pyrolysis route was changed and produced more CO₂ and H₂O. Table 2 shows that the torrefaction pretreatment had a stronger impact on the volume fraction of the non-condensable gases. Compared to the non-condensable gases produced by the pyrolysis of CS,

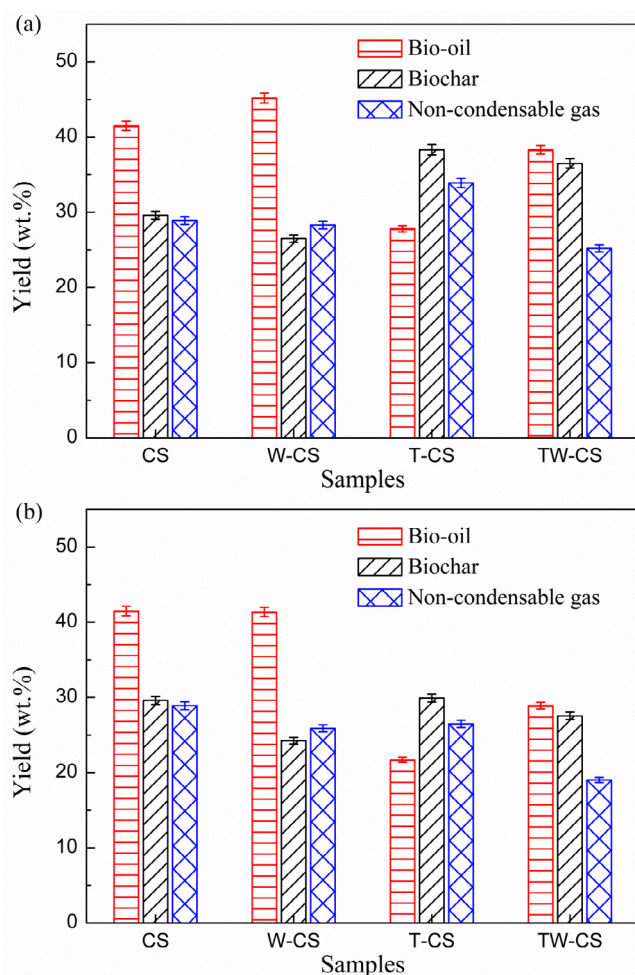


Fig. 4. Yields of products generated from the pyrolysis of CS, W-CS, T-CS, and TW-CS: (a) on the base of the pretreated samples, (b) on the base of the original cotton stalk.

the volume fraction of CO_2 from T-CS decreased, whereas the volume fractions of CO , H_2 , and CH_4 increased. The decarboxylation of hemicellulose during the torrefaction pretreatment produced CO_2 and thus lowered the CO_2 content produced during subsequent pyrolysis reactions. In contrast, the torrefaction pretreatment had little effect on the cellulose and its carbonyl functional groups, and thus the volume fraction of CO derived from decarbonylation changed slightly. During pyrolysis, CH_4 was mainly generated from cleavage of the methoxy group on the benzene ring of lignin, and since the torrefaction pretreatment increased the lignin content, the volume fraction of CH_4 also increased. Compared to the torrefaction pretreatment, the combined pretreatments further increased the volume fractions of CH_4 and H_2 and decreased the volume fraction of CO_2 . All of the pretreatment methods increased the combustible component content, which increased the HHV of the non-condensable gases.

Table 2

The volume fractions and HHV of non-condensable gases issued from pyrolysis of CS, W-CS, T-CS, and TW-CS.

Non-condensable gas	Volume fractions (vol.%)					HHV (MJ/Nm ³)
	CO_2	CO	CH_4	H_2	C_2^+	
CS	43.16 ± 0.55	28.73 ± 0.35	16.28 ± 0.21	9.21 ± 0.16	2.62 ± 0.05	13.45 ± 0.46
W-CS	41.44 ± 0.53	28.91 ± 0.38	17.85 ± 0.25	9.53 ± 0.15	2.27 ± 0.04	13.78 ± 0.43
T-CS	36.92 ± 0.49	30.26 ± 0.41	18.67 ± 0.27	11.28 ± 0.21	2.87 ± 0.03	14.43 ± 0.38
TW-CS	36.36 ± 0.52	29.77 ± 0.36	19.58 ± 0.26	12.16 ± 0.22	2.13 ± 0.04	14.76 ± 0.43

3.4.2. Bio-oil

The water content, HHV, and pH of the bio-oil are listed in Table 3. The bio-oil obtained from CS was 58.19 wt.% water. However, this was decreased by both the aqueous phase bio-oil washing pretreatment and the torrefaction pretreatment. The bio-oil obtained from TW-CS had the lowest water content with a value of 34.45 wt.%. Pretreatment with aqueous phase bio-oil washing removed the metallic species and thereby reduced their catalytic effects on the dehydration reaction and promoted the development of pyrolysis intermediates to form bio-oil. For the torrefied cotton stalk, the intrinsic structure and component content of the biomass changed and led to less water formation. Meanwhile, the pH and HHV of the bio-oil increased with both the aqueous phase bio-oil washing pretreatment and the torrefaction pretreatment. The HHV of the bio-oil increased from 11.34 MJ/kg for CS to a maximum value of 15.95 MJ/kg for TW-CS. These findings indicate that the fuel qualities of the bio-oil were improved by all of the pretreatments and that the combined pretreatments had more effect than the individual pretreatments.

Bio-oil contains hundreds of chemicals. The components identified using GC/MS were generally categorized according to the functional groups of acids, ketones, aldehydes, phenols, furans, and anhydrosugars, and the peak area% are shown in Fig. 5. The sum of the total peak area% was less than 100%, and this may due to the presence of unidentified compounds in the ion chromatograms. Peak area% varies linearly with the content, and thus the changes in the relative content of each product among the detected products can be determined by comparing the peak area% values (Lu et al., 2014).

The bio-oil obtained by pyrolysis of CS had the highest relative acid content, and this was mainly acetic acid. Both of the pretreatment methods significantly reduced the relative acid content in the bio-oil. The single aqueous phase bio-oil washing pretreatment had little effect on the relative content of furans and aldehydes but increased the relative content of phenols. However, the relative content of phenols increased obviously after torrefaction, while the relative content of ketones and furans decreased. These observations were related to the changes in metallic species content and the components of the samples. With the catalysis of metallic species during the pyrolysis of cotton stalks, small molecular products were readily produced. For example, K and Na catalyzed reactions such as ring fission and fragmentation of glucose, decreasing the formation of anhydrosugars while increasing the yield of low molecular weight compounds such as acetic acid, aldehydes, and ketones (Lv and Wu, 2012; Mourant et al., 2011). Torrefaction pretreatment reduced the hemicellulose content and increased the lignin content. As a result, fewer acids were produced during the pyrolysis of T-CS since acetic acid was mainly produced from the pyrolysis of hemicellulose. Meanwhile, more phenols were produced during the pyrolysis of T-CS since phenols were the main liquid products produced from lignin pyrolysis.

It should be noted that anhydrosugars such as L-glucose were detected with a very small peak area% in the bio-oils produced by pyrolysis of all of the samples. These have been otherwise reported in large amounts in biomass pyrolysis using Py-GC/MS

Table 3

The water content, HHV and pH values of bio-oil issued from pyrolysis of CS, W-CS, T-CS, and TW-CS.

Bio-oil	Water content (wt.%)	HHV (MJ/kg)	pH
CS	58.19 ± 0.92	11.34 ± 0.28	2.95 ± 0.06
W-CS	51.07 ± 0.72	13.46 ± 0.23	3.08 ± 0.04
T-CS	41.13 ± 0.65	14.93 ± 0.24	3.12 ± 0.04
TW-CS	34.45 ± 0.38	15.95 ± 0.26	3.13 ± 0.05

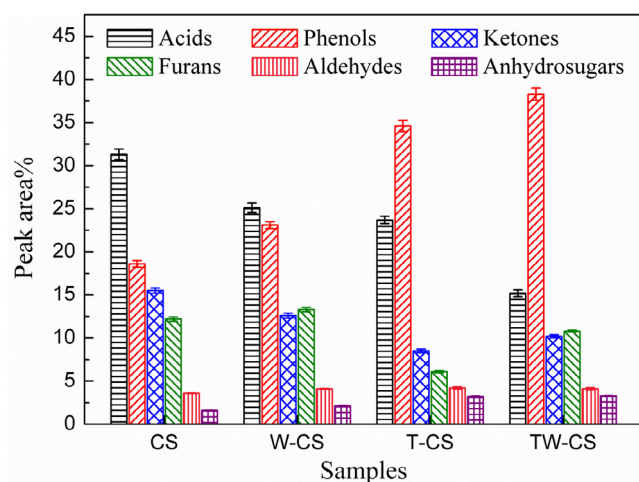


Fig. 5. The relative content of different groups of bio-oil generated from pyrolysis of CS, W-CS, T-CS, and TW-CS.

(Chen et al., 2014b; Lu et al., 2014; Pecha et al., 2015; Zhang et al., 2016b). This might be because of differences in experimental conditions. Characteristics of Py-GC/MS pyrolysis are a rapid heating rate (up to 20 °C/ms), a short reaction time (about 10 s), real-time detection of the pyrolysis products, and the absence of a secondary cracking reaction of the pyrolysis products. In this study, pyrolysis was performed in a fixed bed pyrolysis reactor, and the experiment time was set at 15 min to ensure the collection of gaseous products and complete pyrolysis. The secondary cracking reaction of sugars such as L-glucose is greatly affected by temperature and reaction time. Increasing the reaction time increases the cracking of sugars to produce small molecules such as aldehydes and ketones.

The higher contents of acids and water in bio-oil are the main reasons for the high oxygen content, poor thermal stability, and high acidity. After the aqueous phase bio-oil washing and torrefaction pretreatments, the quality of bio-oil obtained from pyrolysis of cotton stalks improved considerably. The bio-oil derived from the pyrolysis of TW-CS had the lowest water content, the highest HHV, the lowest acid content, and the highest phenol content compared to those obtained from pyrolysis of CS, T-CS, and W-CS.

3.4.3. Biochar

The basic properties of biochar are given in Table 4. After pretreatment with aqueous phase bio-oil washing, the ash content of biochar decreased, whereas the torrefaction pretreatment minimally changed the ash content of biochar. The combined pretreatment resulted in a slight decrease in the ash content of biochar, and this was mainly because some amount of ash was removed by the aqueous phase bio-oil washing before pyrolysis of the cotton stalks. Biochar also contains some metallic species, and this made the pH of the biochar alkaline. In general, both the aqueous phase bio-oil washing and the torrefaction pretreatment did not have significant effects on the proximate analysis, the ultimate analysis, or the HHV of the biochar. In contrast, these pretreat-

ments had clear influences on the pore structure of the biochar. All of the pretreatments, and especially the bio-oil washing pretreatment, increased the specific surface area of biochar, and made biochar amenable for use as an adsorbent or as a catalyst.

3.5. Simple comparison with other pretreatments

At present, the most commonly used pretreatments for improving the quality of biomass and pyrolysis products are water washing and acid washing. Previous studies have reported the effects of water-based or dilute acid-based washing pretreatments on the fuel property and/or pyrolysis characteristics of rice husks, wheat straw, corn stalks, moso bamboo, and Douglas fir wood (Deng et al., 2013; Dong et al., 2015; Pecha et al., 2015; Zhang et al., 2015). Water washing pretreatment of biomass removes some of the metals, such as K, Ca, and Mg. However, since the chemical structure of biomass is not affected by water washing, it has limited influence on the quality of either the biomass or the pyrolysis products. In contrast, acid washing pretreatments using inorganic acids such as hydrochloric acid and sulfuric acid have been used to remove the metallic species in biomass noticeably and to improve the yield and quality of bio-oil (Dong et al., 2015; Ly et al., 2015; Pecha et al., 2015). Nevertheless, such a pretreatment has certain disadvantages, including the unnecessary introduction of elements such as Cl and S into the biomass and the high cost of acid washing (Liu et al., 2013; Zhang et al., 2016a). In addition to inorganic acids, organic acids such as acetic acid have also been used to efficiently remove alkali and alkaline earth metals (Oudenhoven et al., 2013; Pecha et al., 2015). We presented a novel pretreatment that combined torrefaction and washing with torrefaction liquid products to yield upgraded biomass and pyrolysis products (Chen et al., 2017). The torrefaction pretreatment produces small amounts of liquid products called “torrefaction-liquid”, which is mostly composed of water and small amounts of acetic acid. In our previous study, we showed that the combined pretreatment allowed the reuse of liquid products obtained from the torrefaction pretreatment and also improved the quality of the biomass and the pyrolysis products.

In the current study, another novel approach is presented to upgrade the quality of the biomass pyrolysis products using an aqueous phase bio-oil washing and torrefaction pretreatment. A simple comparison between the results of this study and those of the cited studies shows that the effects of the aqueous phase bio-oil washing were better than those with water washing and torrefaction-liquid washing. In particular, there was better removal of metallic species from cotton stalks, an increase in the bio-oil yield, a decrease in the water and acid contents with an increase in the phenol content of the bio-oil, and an improvement in the HHV of the non-condensable gases. The effects of the aqueous phase bio-oil washing are similar to the effects observed with washings based on dilute acid solutions. Nevertheless, further comparative studies are required.

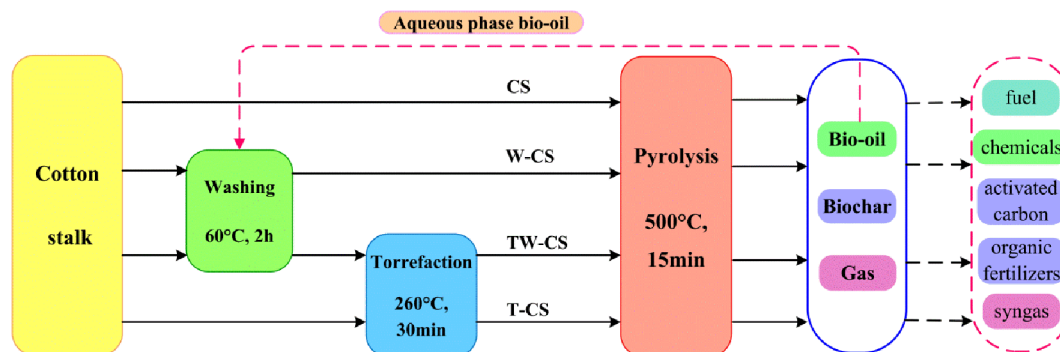
3.6. Schematic flowsheet of pretreatments and pyrolysis polygeneration

The schematic flowsheet of the pretreatments and subsequent pyrolysis procedure used in this study is presented in Fig. 6. In recent years, polygeneration of biomass pyrolysis has attracted wide attention (Chen et al., 2016b; Yang et al., 2016). Polygeneration pyrolysis improves the quality and application of specific pyrolysis products such as bio-oil, and also focuses on using all of the solid, liquid, and gaseous products of pyrolysis along with the waste heat. The non-condensable gases can be used for combustion and heating, and biochar has been widely used as fuel and raw material for preparing catalysts, activated carbon, organic

Table 4

The properties of biochar issued from pyrolysis of CS, W-CS, T-CS, and TW-CS.

Biochar	Proximate analysis (wt.%, db)			Ultimate analysis (wt.%, db)				HHV (MJ/kg)	pH	S _{BET} (m ² /g)
	Volatile	Fixed carbon	Ash	[C]	[H]	[O]	[N]			
CS	26.79 ± 0.41	54.69 ± 0.58	18.52 ± 0.27	70.32 ± 0.63	2.26 ± 0.03	7.58 ± 0.15	1.32 ± 0.02	26.59 ± 0.38	9.02 ± 0.05	134.63 ± 6.73
W-CS	27.32 ± 0.38	59.39 ± 0.52	13.29 ± 0.27	73.56 ± 0.67	2.25 ± 0.03	9.67 ± 0.14	1.23 ± 0.02	27.38 ± 0.26	8.35 ± 0.04	196.45 ± 6.49
T-CS	25.43 ± 0.39	55.51 ± 0.51	19.08 ± 0.29	70.16 ± 0.61	2.21 ± 0.03	7.23 ± 0.14	1.32 ± 0.02	26.12 ± 0.41	9.93 ± 0.04	172.53 ± 7.35
TW-CS	26.75 ± 0.35	58.12 ± 0.63	15.13 ± 0.23	73.65 ± 0.58	2.29 ± 0.03	7.56 ± 0.13	1.37 ± 0.03	27.43 ± 0.44	9.45 ± 0.05	181.29 ± 8.46

**Fig. 6.** Schematic flowsheet of cotton stalk pretreatments and subsequent pyrolysis procedure.

fertilizers, and fast-burning carbon (Chen et al., 2016a; Kambo and Dutta, 2015; Kovacs and Meggyes, 2009). In recent years, the application of biochar for soil amendment and gas storage has received increasing attention because of its excellent adsorption characteristics and stability (Chen et al., 2016c; Qian et al., 2015). Bio-oil is a promising green fuel, and it can be used for direct combustion, chemical extraction after refinement, and preparation of high-value liquid fuel (Chen et al., 2014c; Lehto et al., 2014; No, 2014). At present, bio-oil use is greatly limited because of its low quality, complex composition, high water content, poor thermal stability, and easy phase separation after storage. Since phase separation of bio-oil is usually inevitable, it may be a good idea to study and find uses for the separated phases of bio-oil. The aqueous phase bio-oil contains a large amount of water and acids that can be used to wash the raw biomass and improve its quality. The oil phase bio-oil has a higher HHV and more chemicals that may be used to extract other chemicals or to prepare high value fuel. The above results showed that the aqueous phase bio-oil washing reduced the catalytic pyrolysis of cotton stalks, increased the bio-oil yield, and reduced the water and acid contents of bio-oil. Meanwhile, torrefaction decreased the oxygen content and increased the HHV of cotton stalks; it also improved the HHV of pyrolysis products, reduced the acid content, and increased the phenol content of bio-oil. Generally, the combined pretreatments yielded superior quality gaseous and liquid products after pyrolysis than either of the single pretreatments did, and this indicates that our proposed method is a promising pretreatment, which contributes to the pyrolysis polygeneration of biomass.

4. Conclusion

Aqueous phase bio-oil produced from bio-oil phase separation had large amounts of water and acids. Aqueous phase bio-oil washing pretreatment effectively removed the metallic species from cotton stalks, which further reduced the catalytic effects of metallic species during pyrolysis. In addition, this pretreatment increased the yield of bio-oil and improved the quality of the pyrolysis products (bio-oil, biochar, and non-condensable gases), which also contributed to the pyrolysis polygeneration of biomass. This pretreatment was more effective when combined with the tor-

refaction pretreatment. Therefore, the combination of aqueous phase bio-oil washing and torrefaction is a promising method for biomass pretreatment.

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